

How can the second response be faster?

Learning intention

Explain how barriers and the specific immune response protect the body, including memory after immunisation.

Mission: build a defence explanation from cells, proteins and evidence.

Use the model; no personal health stories needed.

Three distinctions

R1. What is a pathogen?

R2. What does a physical barrier do?

R3. Is a cell the same thing as a protein?

Before a specific response

Defences at the entry points

Skin
Physical barrier

Airway mucus + cilia
Trap material,
then move it

Tears: lysozyme
Damages some
bacterial walls

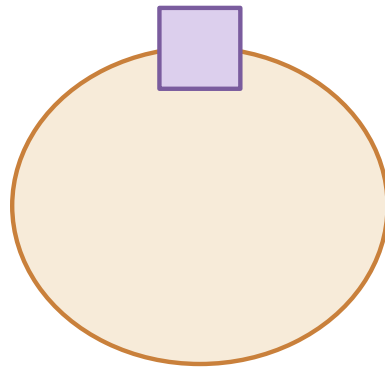
Stomach acid
Kills many swallowed
pathogens

Barriers reduce entry or survival; they are not perfect.

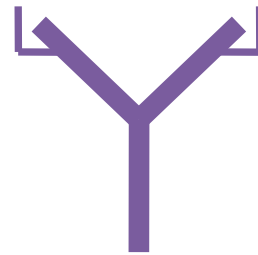
Match a structure with its protective action. I explain how it works, not just where it is.

Recognise the antigen

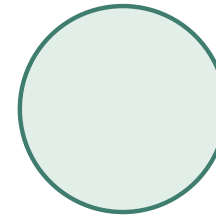
A protein can recognise a specific antigen



Pathogen surface
Antigen X: square



Antibody protein
Binding-site model



Antigen Y
Different fit

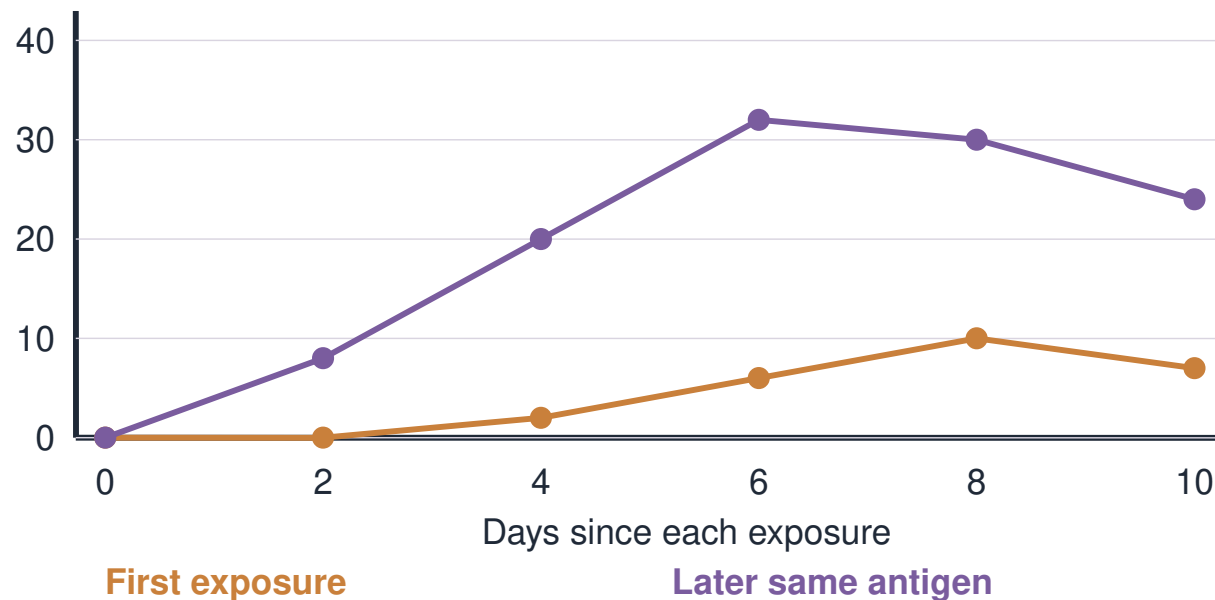
A simplified recognition model; shapes are not to scale.

An antigen can trigger a specific immune response. Antibodies are proteins made by specialised cells. Binding depends on the antigen.

Read the time and amount

Same antigen: compare the responses

Antibody amount (arbitrary units)



At day 4: first response 2 units; later response 20. The later response begins sooner in this model. Memory lymphocytes support the faster response.

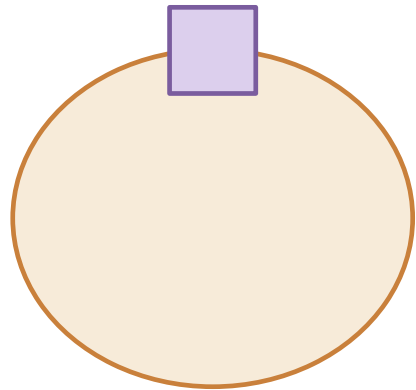
Build the explanation chain

W1. Order: antibody response • antigen exposure • memory retained • faster later response.

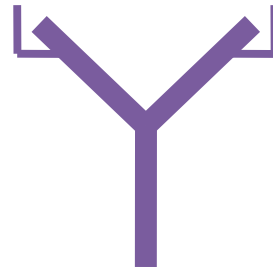
Give each arrow a reason.

Choose a target, test the fit

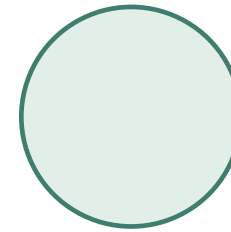
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W2. Would an X-specific antibody necessarily bind Y? Explain what the model can and cannot show.

Evidence desk: compare day 6

W3. Find each day-6 amount and the difference. Explain one reason and one limit.

Days after exposure	First: units	Later same antigen: units
0	0	0
2	0	8
4	2	20
6	6	32
8	10	30
10	7	24

Constructed teaching data; arbitrary units, not protective thresholds.

Which mechanism is correct?

Choose one explanation and defend it.

- A** Antibodies are memory cells.
- B** Memory lymphocytes support a faster response to the same antigen.
- C** One antibody protects equally against every pathogen.
- D** Antibiotics work by creating immune memory.

Create a defence explanation

Choose the agreed response page.

Use an arrow chain, short sentences or a narrated diagram.

Make the model teach someone

Include a barrier, recognition and memory.

Use one numerical comparison from the graph.

Audience test: is the mechanism clear?

L1. Share one part of the explanation.

Audience: “Which is a cell? Which is a protein?”

Improve the final claim

“The vaccine gives perfect protection forever.”

Replace certainty with a careful mechanism.

Explain the faster response

- E 1. Name one barrier and its action.
- E 2. Is an antibody a cell or protein?
- E 3. How does memory help next time?

Choose the next useful step

Barrier action • specific binding • memory • graph reading.